MACHINE LEARNING BASED INTELLIGENT EV BATTERY CHARGING WITH DIFFERENTIAL POWER PROCESSING

SDG MAPPING

SDG Goal		Remarks
SDG 7	Affordable And Clean Energy	Improves battery energy utilisation through DPP-based cell balancing. Reduces unnecessary energy losses during the balancing process.
SDG 9	Industry, Innovation and Infrastructure	Integrates XGBoost, ESP32, and intelligent charging control. Supports innovation in EV battery-management technology.
SDG 11	Sustainable Cities and Communities	Supports reliable EV battery operation through health-aware charging and temperature monitoring. Encourages efficient and sustainable electric mobility.

IEEE PROFESSIONAL STANDARDS

IEEE Standards Number	IEEE Standards Name	Remarks
IEEE 1547-2018	Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces	Relevant to the integration of the EV charging system with distributed energy resources and the electrical grid. Provides guidelines for grid interconnection and interoperability when the charging setup is connected to such resources.
	Practice for the Characterization and Evaluation of	Provides guidance for evaluating energy-storage technologies based on performance, service life, and safety characteristics. Its

IEEE 1679-2020	Energy Storage Technologies in Stationary Applications	principles can inform battery evaluation, although its scope is stationary energy storage rather than EV battery packs specifically.

ABSTRACT

The increasing adoption of electric vehicles (EVs) requires efficient, reliable, and safe battery charging systems. Lithium-ion batteries are widely used in EVs, but cell imbalance, battery ageing, and temperature variations can reduce performance and safety. This project presents a Machine Learning-Based Intelligent EV Battery Charging System using Differential Power Processing (DPP) for active cell balancing and health-aware charging control. DPP transfers energy between adjacent cells by processing only the mismatch power, potentially reducing converter power ratings and improving energy utilization. The Extreme Gradient Boosting (XGBoost) algorithm estimates the State of Health (SoH) and Remaining Useful Life (RUL) using battery voltage, charging current, discharging current, and temperature measurements. A gradient-boosted-tree classifier selects the charging mode based on battery health, State of Charge (SoC), cell voltage imbalance, temperature, and available charging current. An ESP32 microcontroller manages charging control, cell balancing, and protection, while a Wi-Fi-based dashboard supports remote monitoring. The system operates in four modes: balance-priority, SoH-derated Constant Current–Constant Voltage (CC-CV), fast charging, and protection-and-monitoring. MATLAB/Simulink modelling, analytical calculations, and numerical simulations evaluate the balancing performance and intelligent charging strategy. In the four-cell simulation, the SoC spread decreases from 10% to 2% within 110 minutes. The analysis also demonstrates that charging current decreases as battery health deteriorates, helping reduce stress on aged cells.

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LIST OF ABBREVIATIONS

S.NO	ABBREVIATION	FULL DESCRIPTION
1	EV	Electric Vehicle
2	ML	Machine Learning
3	AI	Artificial Intelligence
4	DPP	Differential Power Processing
5	BMS	Battery Management System
6	SoC	State of Charge
7	SoH	State of Health
8	RUL	Remaining Useful Life
9	XGBoost	Extreme Gradient Boosting
10	Li-ion	Lithium-Ion
11	CC–CV	Constant Current–Constant Voltage
12	ESP32	ESP32 Microcontroller
13	PWM	Pulse Width Modulation
14	DC	Direct Current
15	MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
16	ADC	Analog-to-Digital Converter
17	Wi-Fi	Wireless Fidelity
18	MATLAB	Matrix Laboratory
19	MAE	Mean Absolute Error
20	RMSE	Root Mean Square Error

CHAPTER 1

INTRODUCTION

The increasing adoption of electric vehicles (EVs) has created a need for battery charging systems that provide efficient, reliable, and safe operation. The battery pack is one of the most important components of an electric vehicle because it determines the vehicle's driving range, charging time, and overall performance. Lithium-ion batteries are widely used in EVs because of their high energy density, good efficiency, and long cycle life. However, variations in cell characteristics, temperature, and ageing can affect battery performance. Therefore, an intelligent battery charging system is required to monitor individual cells, maintain cell balance, and adjust charging according to battery health. The system aims to improve cell balancing and charging management by considering the condition of individual battery cells rather than depending only on fixed charging parameters. Machine learning is used to estimate battery health and remaining useful life, while DPP enables energy transfer between cells with different states of charge. An ESP32-based controller is considered for monitoring and control, and MATLAB/Simulink is used to model and evaluate the proposed system.

1.1 BACKGROUND OF THE WORK

An electric vehicle battery pack generally consists of multiple cells connected in series and parallel to obtain the required voltage and capacity. In a series-connected battery pack, the same current flows through each cell, but the cells may have different capacities, internal resistances, temperatures, and states of charge. These variations can cause cell imbalance during charging and discharging. If one cell reaches its upper voltage limit before the other cells, charging may need to be reduced or stopped to protect the battery. Similarly, a cell with a lower state of charge or reduced capacity can affect the usable energy of the entire battery pack. A battery management system (BMS) monitors important battery parameters such as cell voltage, current, temperature, and state of charge. It also provides protection against abnormal operating conditions. Conventional charging systems commonly

use the constant-current–constant-voltage (CC-CV) method. During the constant-current stage, the battery is charged with a controlled current until it reaches the specified voltage limit. During the constant-voltage stage, the voltage is maintained while the charging current gradually decreases. Although this method is widely used, a fixed charging strategy may not fully account for differences in cell health, temperature, and imbalance. Consequently, charging performance and battery protection can be affected when the cells have different operating conditions.

Cell balancing is an important function in battery management. Passive balancing removes excess energy from higher-voltage cells through resistive elements, converting the excess energy into heat. Although this approach is relatively simple, it results in energy loss. Active balancing transfers energy from cells with higher energy levels to cells with lower energy levels, improving energy utilisation. Differential power processing is an active balancing approach in which power converters transfer only the required mismatch power between cells instead of processing the entire battery-pack power. In the proposed system, differential power processing converters are connected between adjacent cells to support energy transfer and reduce differences in their states of charge. Machine learning can further improve battery management by analysing measured operating parameters and estimating battery condition. State of charge (SoC) indicates the remaining charge available in a battery, while state of health (SoH) represents its condition relative to its original performance. Remaining useful life (RUL) estimates the remaining service life, commonly expressed in terms of expected cycles or operating duration. The proposed system uses an XGBoost model to estimate SoH and RUL from parameters such as terminal voltage, charging current, discharging current, and temperature.

1.2 MOTIVATION AND SCOPE OF THE WORK

The main motivation of this project is to address the limitations of conventional EV battery charging systems, particularly energy loss during passive balancing, unequal cell charging, and the use of charging strategies that do not adequately

consider battery health. A battery pack may contain cells with different states of charge and ageing characteristics. Applying the same charging conditions without considering these differences may reduce charging flexibility and increase the possibility of reaching cell voltage or temperature limits. Therefore, a coordinated approach combining active cell balancing, machine learning, and real-time monitoring is proposed. The project focuses on developing and evaluating an intelligent charging architecture using differential power processing and machine-learning-based battery assessment. DPP converters are intended to transfer mismatch energy between adjacent cells, helping to reduce cell imbalance. The XGBoost model estimates SoH and RUL using relevant battery operating parameters. Based on battery health, cell imbalance, and temperature, a gradient-boosted-tree classifier supports the selection of suitable charging operating modes. The proposed control strategy includes four operating modes. Balance-priority mode gives preference to cell balancing when the maximum cell-voltage difference exceeds the specified threshold. SoH-derated CC-CV mode adjusts charging current according to battery health and voltage conditions. Fast-charge mode is considered when the battery satisfies the specified health, temperature, and cell-balance conditions. Protection and monitoring mode responds to faults or excessive temperature by disabling charging and isolating the battery as required by the control strategy. These modes are intended to coordinate charging performance with battery safety.

An ESP32 microcontroller is considered for controller implementation, measurement processing, and pulse-width modulation (PWM) generation for converter control. The proposed architecture also includes a Wi-Fi-based dashboard for displaying relevant battery information and supporting remote monitoring. MATLAB/Simulink is used to model the multi-cell battery pack, cell-balancing converters, measurement signals, and switching-control logic. The model allows the proposed control strategy and balancing behaviour to be examined under defined operating conditions before hardware implementation.

Summary

The proposed intelligent EV battery charging system integrates Differential Power Processing (DPP), machine-learning-based battery health estimation, and adaptive charging control to improve battery management. DPP converters facilitate energy transfer between adjacent cells to reduce SoC imbalance, while the XGBoost model estimates SoH and RUL using battery operating parameters. The system incorporates four charging modes, temperature monitoring, and fault protection to support safe and efficient operation. An ESP32-based controller and Wi-Fi dashboard are considered for monitoring and control, while MATLAB/Simulink is used for modelling and simulation. The study evaluates the proposed approach through analytical calculations and numerical simulations, with hardware implementation and experimental validation identified as future work.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

The increasing adoption of electric vehicles has created a need for efficient battery-management systems. Lithium-ion battery packs may experience cell imbalance, ageing, and temperature variations that affect performance and safety. Existing studies have investigated active cell balancing, Differential Power Processing (DPP), machine-learning-based health estimation, and intelligent charging strategies. Reviewing these approaches helps identify limitations in existing methods and opportunities for improvement. This chapter examines relevant research and identifies the gap addressed by the proposed intelligent EV battery charging system.

1. Ghaemi Nezhad et al. (2021) studied active cell equalization topologies for lithium-ion battery packs. The researchers classified the balancing methods into adjacent-based, non-adjacent-based, direct cell-to-cell, and mixed topologies. Mathematical models and simulation results were used to compare the equalization speed and economic aspects of different balancing techniques. The study provided a systematic understanding of active balancing methods and their suitability for battery packs. However, the review did not focus on integrating machine-learning-based battery health estimation with intelligent charging control. This provides a basis for the proposed system, which combines DPP-based balancing with battery health monitoring [1].

2. Krishnamoorthy et al. (2021) investigated the performance of active cell balancing for series-connected lithium-ion batteries used in electric vehicles. The study focused on the importance of transferring energy between cells to reduce differences in their states of charge. Active balancing was considered as a method to improve battery-pack utilisation and address cell imbalance. The work highlights the importance of selecting a suitable balancing topology for series-connected battery packs. However, integrating battery State of Health (SoH)

estimation and intelligent charging-mode selection with the balancing system remains an important area for further investigation [2].

3. Manjunath et al. (2021) proposed a two-stage, module-based cell-to-cell active balancing circuit for series-connected lithium-ion battery packs. The topology used a modified buck-boost converter to transfer energy between cells and support active equalisation. The study examined the circuit structure and its suitability for balancing cells in a battery pack. This approach is relevant to electric vehicles because cell imbalance can limit the usable capacity of the complete pack. However, converter operation alone does not provide battery health prediction or adaptive charging decisions. The proposed project extends this approach by combining active balancing with machine-learning-based SoH estimation [3].

4. Matrone et al. (2021) developed a machine-learning-based technique for classifying electric-vehicle charging sessions to improve charging-duration prediction. The researchers used the K-Nearest Neighbours (KNN) algorithm and considered information available at the beginning of a charging session, including arrival time, initial State of Charge (SoC), and calendar data. The model was evaluated using real-world charging-session data collected from more than 100 users, and its performance was compared with benchmark models. The study demonstrated the usefulness of machine learning for predicting charging behaviour. However, its primary focus was charging-session duration rather than cell-level balancing and battery SoH estimation. The proposed project applies intelligent control to battery balancing and health-aware charging decisions [4].

5. Khan et al. (2022) investigated the estimation of lithium-ion battery State of Health using an XGBoost-based method combined with an adaptive Kalman filter. The study addressed the need to estimate battery degradation accurately for energy-storage applications. Combining machine learning with filtering techniques provides an approach for estimating battery health from operating data. Such estimation can support battery monitoring and maintenance decisions.

However, health estimation must be connected to the charging controller to influence actual charging decisions. The proposed project uses XGBoost-based SoH estimation as an input for selecting suitable charging modes and balancing priorities [5].

6. Xiao et al. (2022) studied lithium-ion battery State of Health estimation using an explainable XGBoost model with parameter optimisation. The research addressed the importance of obtaining accurate battery health estimates while improving the interpretability of machine-learning predictions. The XGBoost approach can identify relationships between battery operating features and health indicators. This is relevant to battery management systems, where health information can support charging and protection decisions. However, accurate prediction alone does not balance individual cells or determine the appropriate charging mode. The proposed system integrates battery health estimation with DPP-based cell balancing and intelligent charging control [6].

7. Liu et al. (2023) investigated adaptive State of Health estimation for lithium-ion batteries using partially labelled and incomplete charging curves. The study addressed the practical difficulty of estimating battery health when complete charging data are unavailable. An adaptive estimation approach can be useful when battery operating conditions and available measurements vary. This research is relevant to real-world battery management, where continuous and complete data may not always be accessible. However, the integration of adaptive health estimation with cell-level energy balancing and temperature-based charging protection requires further development. The proposed project aims to combine health-aware control with active balancing and temperature monitoring [7].

8. Su and Li (2024) studied a differential-power-delivery-based control framework for multiple battery energy-storage systems in DC microgrids. The work considered how power can be distributed among storage units to support coordinated operation. Differential power delivery is relevant to energy-management applications because it can provide a way to control power transfer rather than treating all storage elements identically. However, a system-level DC

microgrid differs from balancing individual lithium-ion cells within an EV battery pack. The proposed project applies the differential power-processing concept to cell-level balancing and combines it with battery health estimation and intelligent charging strategies [8].

9. Khan et al. (2024) presented an active cell-balancing topology for series-connected lithium-ion cells in electric-vehicle battery packs. The study addressed the problem of cell-to-cell imbalance, which can reduce the effective utilisation of a battery pack. Active balancing transfers energy between cells instead of relying only on dissipating excess energy through resistive elements. This approach is relevant to improving the utilisation of cells with different SoC values. However, a balancing circuit needs suitable monitoring and control strategies to respond to changes in battery health and temperature. The proposed system integrates DPP-based balancing with machine-learning-based health estimation and charging-mode selection [9].

10. Venkatasatish and Dhanamjayulu (2024) examined an inductor-based approach to active battery-cell balancing. Inductor-based balancing circuits use energy-storage elements to transfer energy between cells and reduce differences in their charge levels. The selection of inductance and switching parameters is important because these values influence current transfer and balancing performance. This work is relevant to the design of a converter-based balancing system for EV batteries. However, converter design must also be coordinated with battery health, temperature, and charging-current limits. The proposed project considers converter parameters together with intelligent charging control to improve the overall battery-management strategy [10].

11. Praveena Krishna et al. (2024) investigated a switched-supercapacitor-based approach to active cell balancing. Supercapacitors can temporarily store energy transferred from one cell and deliver it to another, helping to reduce differences in cell charge levels. The study is relevant to the design of balancing circuits because the selection of energy-storage components affects circuit complexity and energy-transfer behaviour. However, switched-supercapacitor methods require

appropriate switching and control arrangements, and their depends on the selected topology. The proposed project instead focuses on DPP-based and combines it with machine-learning-based battery health estimation and adaptive charging modes [11].

12. Sudhapriya and Jaisiva (2025) reviewed the application of artificial intelligence in battery-management systems. Artificial-intelligence techniques can support the analysis of battery operating data and help estimate parameters such as SoC, SoH, and temperature-related behaviour. These techniques can assist in monitoring battery condition and making data-driven control decisions. However, the practical benefits depend on data quality, model accuracy, and the integration of predictions with the battery-management controller. The proposed project focuses on connecting machine-learning-based battery health estimation with DPP balancing and intelligent charging strategies [12].

13. Pujari Harish Kumar et al. (2025) examined battery-management and thermal-management considerations relevant to battery operation. Temperature monitoring is important because battery charging and discharging under unsuitable thermal conditions can affect performance, ageing, and safety. A battery-management system must therefore consider operating temperature alongside electrical parameters. However, temperature monitoring alone does not eliminate cell imbalance or provide a complete battery-health assessment. The proposed temperature-based protection with SoH estimation, DPP-based balancing, and intelligent charging-mode selection to provide a more integrated battery-management approach [13].

14. Sreedharan et al. (2025) investigated the estimation of lithium-ion battery SOH using an XGB-AKF method for energy-storage applications. The approach combines XGBoost with adaptive Kalman filtering to support battery-health estimation. Machine learning can identify relationships between battery measurements and degradation, while filtering can help process changing measurements. The study provides a useful reference for developing data-driven

battery-health estimation methods. However, the estimated SOH must be connected to a suitable charging-control strategy to adjust operating conditions according to battery ageing and condition [14].

15. Njoku et al. (2025) reviewed the use of digital-twin technology for battery management. A digital twin can represent a physical battery system using a computational model updated with relevant operating data. This approach can support monitoring, simulation, performance analysis, and maintenance planning. Digital-twin methods are relevant to intelligent battery systems because they can connect battery behaviour with computational analysis. However, a digital representation must be supported by reliable data and validated models to reflect the physical battery accurately. The proposed project uses MATLAB/Simulink simulation to study cell balancing and charging control, with hardware integration and real-time monitoring identified as future development activities [15].

Summary

The reviewed studies focus on various aspects of electric vehicle battery management, including active cell balancing, Differential Power Processing (DPP), machine-learning-based battery health estimation, intelligent charging control, and thermal management. Several researchers have investigated active balancing topologies to reduce State of Charge (SoC) differences between cells and improve battery-pack utilisation. Other studies have explored XGBoost-based techniques for estimating battery State of Health (SoH), while adaptive filtering methods support health estimation under varying operating conditions. Machine-learning-based charging optimisation and digital-twin technologies have also been examined for battery monitoring, performance evaluation, and predictive maintenance. Furthermore, thermal management has been identified as an important factor in maintaining battery performance and supporting safe operation. Despite these developments, integrating cell balancing, battery health prediction, and adaptive charging control within a unified system remains an important research area. Conventional balancing approaches may not fully consider battery ageing, temperature variations, and individual cell conditions

when determining charging strategies. Temperature monitoring and intelligent charging-mode selection are incorporated to support battery protection and health-aware charging decisions. The approach aims to improve cell balancing and charging management, while hardware implementation and experimental validation remain necessary to confirm practical performance, efficiency, reliability, and safety.

CHAPTER 3

OBJECTIVES AND METHODOLOGY

The proposed system integrates differential power processing (DPP), machine-learning-based battery-condition estimation, and health-aware charging control for an electric vehicle (EV) battery pack. The methodology is organised around a four-cell battery model, an XGBoost-based estimator for State of Health (SoH) and Remaining Useful Life (RUL), a charging-mode selection strategy, and converter control. MATLAB/Simulink is used to represent the battery pack, DPP converters, measurement signals, and control logic under specified operating conditions. An ESP32-based controller and Wi-Fi monitoring interface are considered in the proposed architecture. The procedure begins by defining the battery parameters and initial cell conditions. Operating variables, including terminal voltage, charging current, discharging current, and temperature, are then used for battery-condition assessment. The estimated health information is considered alongside cell imbalance and safety conditions to determine the appropriate operating mode. Depending on the conditions, the system prioritises cell balancing, applies SoH-derated constant-current–constant-voltage (CC-CV) charging, permits fast charging when the eligibility conditions are satisfied, or activates protection and monitoring.

3.1 Objectives

The objectives are defined around the practical development and evaluation of the proposed battery charging architecture. The design combines cell-level energy management with data-driven battery assessment so that charging decisions can consider both present operating conditions and estimated battery health.

Development of a DPP-Based Battery Charging System

To develop a battery charging architecture that incorporates differential power processing for active balancing of series-connected cells. Individual cells can develop different states of charge because of variations in capacity, internal resistance, temperature, and ageing. These differences can restrict the usable capacity of the battery pack when one cell reaches its voltage limit before the others. The proposed architecture uses converters between adjacent cells to transfer mismatch energy. For the four-cell configuration considered in the project, three converters are used in an N−1 arrangement. The objective is to evaluate whether this arrangement can reduce cell imbalance under the selected simulation conditions.

3.1.1 Estimation of Battery SoH and RUL

To develop an XGBoost-based estimation procedure for battery SoH and RUL using terminal voltage, charging current, discharging current, and temperature as input features. SoH represents battery condition relative to its original performance, whereas RUL estimates the remaining service life under a defined measure, such as remaining cycles. The procedure includes input-data preparation, model training using reference values, prediction, and evaluation on separate test data. The predicted health information is used as an input to the charging-mode selection strategy. Model performance is assessed using suitable error metrics when reference values are available.

Monitoring of Battery Operating Parameters

To monitor the parameters required for cell balancing, charging control, and protection. These include individual cell voltage, charging current, discharging current, temperature, and SoC information. The parameters provide the information required to assess cell imbalance, estimate battery condition, and determine whether charging can continue under the selected operating mode. In the simulation model, the signals are obtained from the battery and converter models or supplied as defined inputs.

3.1.2 Reduction of Cell Imbalance

To evaluate active cell balancing using DPP converters and reduce the difference in SoC among the cells. The system assesses cell-voltage difference and activates balance-priority operation when the specified condition is met. The balancing performance is evaluated by recording individual cell SoC values and the maximum and minimum values throughout the simulation. The effect of converter inductance and duty ratio on balancing current and balancing time is also examined.

Development of Health-Aware Charging Control

To develop a charging strategy that considers estimated SoH, cell imbalance, and temperature rather than relying exclusively on fixed charging parameters. The strategy combines CC-CV charging with mode-selection logic to determine when balancing, health-adjusted charging, or fast charging should be prioritised. The fast-charge eligibility conditions specified in the project paper include SoH of at least 90% and temperature not exceeding 40°C, subject to the other cell-balance and safety conditions. These thresholds are specific to the proposed model and are not universal limits for all battery chemistries.

Integration of Monitoring and Protection Functions

To incorporate monitoring and protection logic for abnormal battery conditions. A detected fault or excessive temperature must override normal charging modes. The protection response is intended to disable charging and isolate the battery as required by the system design. The proposed architecture considers an ESP32 controller and Wi-Fi-based monitoring interface. These elements are intended to support embedded control and display relevant battery information.

3.2 Synthetic Procedure and Flow Diagram

The proposed procedure consists of configuring the battery model, preparing the operating data, estimating battery health, evaluating cell imbalance, selecting the charging mode, controlling the DPP converters, and evaluating system performance. These stages are connected so that the charging decision responds to the battery's estimated condition and current operating parameters. The procedure combines the modelling and estimation methods with the control

actions required by the proposed. Active cell-balancing methods use power-conversion circuits to redistribute energy between cells rather than dissipating all excess energy through resistors. Different circuit arrangements have different requirements for converter count, switching control, efficiency, and implementation complexity (Ghaeminezhad et al., 2021). The present method uses the DPP arrangement described in the project paper and evaluates its operation with a four-cell battery model.

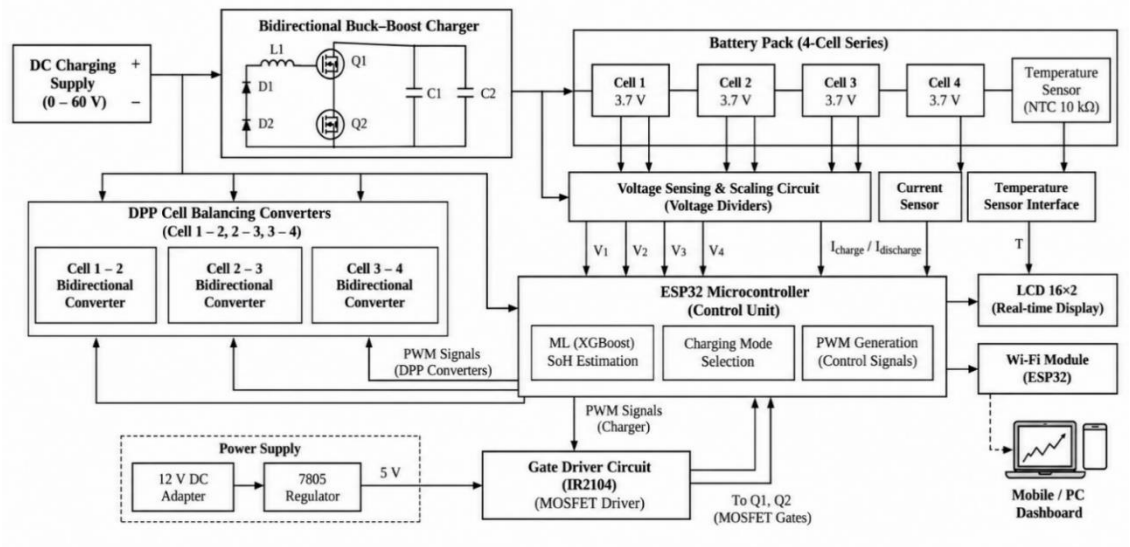


Figure 3.1 Block Diagram

Figure 3.1 illustrate the system uses a bidirectional buck–boost charger to charge a four-cell lithium-ion battery pack. Differential Power Processing (DPP) converters transfer energy between adjacent cells to reduce SoC imbalance. Voltage, current, and temperature sensors send battery data.

3.2.1 Battery-Pack Configuration

A four-cell battery-pack model is used to examine cell imbalance and active balancing. The initial SoC values specified in the project paper are 88%, 84%, 81%, and 78%. These unequal values provide the starting conditions for investigating how the balancing controller responds to differences between cells. The model considers a cell-voltage range of 3.6–4.2 V and an example capacity of 2.5 Ah. These values define the simulation conditions described in the project paper. The model is configured with the required electrical parameters, initial states, and charging conditions. The battery model must provide the cell voltage,

current, and SoC signals needed by the control algorithm.

Data Acquisition and Preparation

The input-data stage provides the operating information required for estimation and control. The features identified in the project paper include terminal voltage, charging current, discharging current, and temperature. Individual cell-voltage and SoC values are additionally required for assessing imbalance and evaluating the balancing response. For simulation-based evaluation, these values are obtained from the battery model or supplied as predefined input conditions. If a separate dataset is used to train the machine-learning estimator, it must include suitable feature values and reference targets for SoH and RUL. The source of the data, the target definitions, the units, and the operating range must be documented.

The preparation process includes checking data consistency, identifying missing or invalid entries, confirming measurement units, and separating training and testing records. Test records must not be used to fit the model if they are intended to provide an independent evaluation. When measured data are used, the report should describe the measurement procedure and sensor limitations.

3.2.2 SoH and RUL Estimation

The XGBoost estimator receives the selected operating features and generates battery-health predictions. SoH indicates the condition of a battery relative to its original performance, while RUL represents its expected remaining service life under a defined measure. The two quantities provide different information and should not be treated as interchangeable. The estimation procedure consists of selecting the features, preparing the reference targets, training the model, producing predictions, and evaluating the outputs. If SoH and RUL are estimated using separate models, each model's inputs and outputs must be specified. The RUL target should also state whether it is measured in remaining cycles or another unit. Model evaluation can include mean absolute error (MAE), root mean square error (RMSE), and a plot comparing predicted and reference RUL values. These measures should be reported only when the required reference values are available. The project literature also discusses

XGBoost-based SoH estimation and the importance of feature selection and model validation (Liu et al., 2024, as described in the supplied literature review).

Cell-Imbalance Assessment

Cell imbalance is assessed by comparing the voltage and SoC values of individual cells. The maximum cell-voltage difference is calculated as

$$\Delta V_{max} = V_{max} - V_{min} \quad (3.1)$$

where V_{max} is the highest cell voltage and V_{min} is the lowest cell voltage. The project paper uses a 30 mV cell-voltage difference threshold to trigger balance-priority operation. If the measured or simulated voltage difference exceeds this threshold, the control strategy prioritises balancing and limits fast charging according to the designed logic. If the difference is within the threshold, the controller evaluates SoH, temperature, and other eligibility conditions before selecting the charging mode. Voltage difference is a practical control input, but it is not identical to SoC difference. Cell voltage can also change with current, temperature, and cell characteristics. Accordingly, the evaluation should compare both cell-voltage difference and SoC spread when the model provides these variables. This makes it possible to determine whether the balancing process is reducing the initial difference in cell conditions.

3.2.3 DPP Converter Control

The DPP converter stage transfers mismatch energy between adjacent cells. For four cells, the project paper specifies three converters under the N−1 configuration. The converter models are connected to the battery cells so that the energy-transfer and balancing responses can be examined during simulation. The specified switching frequency is 50 kHz. Inductance values of 100 μH, 10 μH, and 5.35 μH are considered, with a duty ratio of 0.30 used in one design example. These parameters provide a basis for examining the relationship between converter design and balancing performance.

PWM signals represent the converter switching commands. The control logic determines when balancing is required and provides the relevant command to the converter model. The direction of energy transfer must match the topology and the relative energy levels of the connected cells.

Charging Control and Protection

The charging controller coordinates CC-CV operation with the balancing and battery-health functions. During constant-current charging, the current is maintained at the selected limit until the defined voltage condition is reached. During constant-voltage charging, the voltage is maintained while the current decreases according to the charging model. The health-aware strategy adjusts the charging demand based on estimated SoH and the relevant operating constraints. The controller also considers cell-voltage and temperature conditions. The objective is to coordinate charging with the battery's while avoiding operation beyond the limits specified by the model.

3.3 Selection of Components, Tools, Techniques, Procedures and Testing Methods

The selection of components and tools is based on the requirements of the proposed battery charging architecture. The system requires a battery-pack model, cell-balancing converters, operating-parameter inputs, a battery-health estimator, charging-mode selection logic, and a simulation environment. An ESP32 controller and Wi-Fi monitoring interface are considered for the intended embedded implementation. The following subsections explain the role of each selected element and the procedures used to assess its suitability.

Battery-Pack Model

A four-cell battery-pack model is selected to represent a series-connected arrangement with unequal initial SoC values. A multi-cell model is necessary to examine cell imbalance and the energy transfer between adjacent cells. The four-cell arrangement also provides a direct basis for modelling three DPP converters in the N−1 configuration. The model uses the cell-voltage range and capacity described in the project paper, together with the specified initial SoC values.

3.3.1 DPP Converter Selection

DPP is selected to support active energy transfer between cells. Active balancing methods can reduce the energy wasted in resistive balancing, but their performance depends on topology, component count, converter efficiency, and

control complexity (Ghaeminezhad et al., 2021, as cited in the supplied literature review). The project uses three converters for four cells. A switching frequency of 50 kHz and inductance values of 100 μH, 10 μH, and 5.35 μH are considered in the model. A duty ratio of 0.30 is used in a specified design example.

ESP32 Controller

The ESP32 is considered for embedded monitoring, control-signal generation, and communication. Its intended role is to receive the required battery information, process the charging-mode logic, and provide the relevant commands to the converter-control stage. Its communication capability is also suitable for the proposed Wi-Fi monitoring function. The selected ESP32 variant and peripheral configuration must be checked against the required sampling rate, PWM timing, and communication requirements. Driver circuits, isolation, and hardware protection may be necessary depending on the converter topology and battery voltage. The controller should not be treated as a substitute for independent electrical protection. Unless the physical controller has been connected and tested, its role should be described as part of the proposed architecture rather than as a completed experimental implementation.

3.3.2 Voltage, Current and Temperature Inputs

Voltage, current, and temperature are selected because they support battery-state assessment and control. Cell voltage is used to calculate the maximum voltage difference, current represents charging or discharging conditions, and temperature provides information about the thermal state. The voltage-difference threshold of 30 mV is used in the proposed balance-priority logic. The temperature threshold of 40℃ is used as part of the fast-charge eligibility condition specified in the project paper. These values are model-specific and require appropriate justification before use with a different battery or physical prototype. In simulation, the signals are obtained from the model or supplied as defined inputs. In hardware, sensor selection must account for the expected voltage and current ranges, measurement accuracy, calibration, response time, isolation, and electrical.

XGBoost for SoH and RUL Estimation

XGBoost is selected to estimate SoH and RUL from battery operating data. The algorithm uses an ensemble of decision trees to learn relationships between the input features and target variables. Its use in the proposed architecture is consistent with the supplied literature review, which describes XGBoost-based battery-health estimation and the need for validation of the selected model and features (Liu et al., 2024). The input features are terminal voltage, charging current, discharging current, and temperature. The training process requires reference SoH and RUL values. Training and test data must be separated, and the evaluation should use records that were not used to fit the model. MAE and RMSE can be used to assess prediction error where reference values are available. Actual-versus-predicted plots can also be used to inspect the agreement between predictions and reference targets.

Charging-Mode Classification

The proposed system uses a gradient-boosted-tree classification approach to support charging-mode selection. The four modes are balance-priority, SoH-derated CC-CV, fast charge, and protection and monitoring. Their selection depends on the available health estimates, cell imbalance, temperature, and fault status. The implementation should specify the input features, mode labels, training procedure, and validation method if a trained classifier is used. If the conditions are implemented using explicit decision rules, these rules should be documented separately. This avoids confusing a machine-learning prediction with a fixed safety constraint. The classification method is evaluated by applying representative conditions and comparing the selected mode with the expected response. Protection conditions must override normal charging modes. A trained classifier should not be permitted to authorise charging when mandatory voltage, temperature, or fault conditions are violated.

3.3.3 MATLAB/Simulink Environment

MATLAB/Simulink is selected to model the battery pack, DPP converters, measurement signals, and charging-control logic. Its block-based structure allows the major functions to be represented separately and connected into an integrated model. The battery model provides the cell-state variables, the converter model

represents switching and energy transfer, and the control blocks determine the charging mode and switching commands. The simulation is configured using the parameters described in the project paper. Outputs are recorded for comparison across different initial conditions and converter settings. The simulation environment supports repeatable evaluation under defined assumptions. Nevertheless, the validity of the results depends on the battery model, converter representation, numerical settings, and losses included.

Testing and Evaluation Procedures

Testing and evaluation are performed to assess the performance of the proposed intelligent EV battery charging system under defined simulation conditions. The evaluation focuses on cell balancing, machine-learning estimation accuracy, charging-mode selection, converter performance, and protection response.

Cell-balancing test

The SoC values of the four cells are recorded during simulation. The difference between the maximum and minimum SoC is calculated to evaluate the reduction in cell imbalance. Cell-voltage differences are also monitored to verify the balance-priority control condition.

Converter-parameter test

Different inductance values (100 μH, 10 μH, and 5.35 μH) and selected duty ratios are evaluated. The resulting balancing current and balancing time are compared to understand the influence of converter parameters on balancing performance.

Machine-learning evaluation

The XGBoost model's predicted SoH and RUL values are compared with reference values when available. Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and actual-versus-predicted plots can be used to assess estimation accuracy.

Charging-mode test

Different SoH, cell-voltage difference, and temperature conditions are applied to verify the selection of balance-priority, SoH-derated CC-CV, and fast-charge modes.

Protection test

Simulated fault and overtemperature conditions are introduced to verify that charging is disabled according to the defined protection logic.

Model-consistency test

Cell-voltage limits, current direction, converter switching signals, and charging transitions are checked for consistency with the design assumptions.

Summary

The proposed methodology defines the objectives, model configuration, machine-learning procedure, DPP converter control, charging-mode selection, and performance-evaluation methods for the intelligent EV battery charging system. The four-cell model provides the basis for evaluating cell imbalance, while XGBoost is used to estimate SoH and RUL from selected operating features. The charging strategy coordinates balance-priority operation, SoH-derated CC-CV charging, fast charging, and protection. The chapter also describes the selection of converter parameters, the intended ESP32 controller, the MATLAB/Simulink environment, and the proposed monitoring interface. Testing procedures are established for cell-balancing response, estimation accuracy, mode selection, and protection behaviour. The outputs obtained from these procedures provide the basis for analysing the system's performance in the subsequent results and discussion.

CHAPTER 4
PROPOSED SYSTEM

The increasing adoption of electric vehicles (EVs) has created a need for efficient battery charging systems that can provide reliable operation, improved battery utilisation, and effective battery management. Lithium-ion batteries are widely used in EVs because of their high energy density, relatively low self-discharge rate, and suitability for repeated charging and discharging. However, the performance of a battery pack depends on the condition of individual cells, operating temperature, charging current, and differences in the State of Charge (SoC) and State of Health (SoH) of the cells. If these differences are not managed properly, the usable capacity of the battery pack may be reduced, charging may become less efficient, and individual cells may experience increased stress. A conventional battery charging system generally uses a fixed charging strategy and a Battery Management System (BMS) to monitor operating conditions. Although conventional BMS functions provide essential protection, fixed charging strategies may not fully account for changes in battery health or differences between individual cells.

4.1 Proposed Work

Overview of the Proposed System

The proposed system integrates four major functions: battery monitoring,

machine-learning-based health estimation, DPP-based cell balancing, and intelligent charging control. These functions operate together to improve the management of a multi-cell EV battery pack. The battery pack is represented by four lithium-ion cells connected in series. The initial SoC values used in the proposed model are 88%, 84%, 81%, and 78%. These unequal values represent a condition in which the cells have different available charge levels. When cells with different SoC values are charged together, the higher-SoC cells may approach their voltage limits before the lower-SoC cells are fully charged. Therefore, balancing the cells is important for improving the utilisation of the complete battery pack.

4.1.1 Components and Tools Used

The main components and tools considered in the proposed work are described below.

Lithium-ion battery model

The battery pack consists of four series-connected cell models. The initial SoC values are set to 88%, 84%, 81%, and 78% to represent an imbalanced condition. The cell model provides voltage and SoC information for evaluating charging and balancing behaviour.

Voltage and current monitoring

Cell voltage is required to identify differences between cells and to determine whether balancing should be activated. Charging and discharging current provide information about battery operation and are used as inputs for battery health estimation.

Temperature monitoring

Temperature is considered in the charging decision and protection logic. Monitoring temperature is important because charging under unsuitable thermal conditions may increase battery stress and create safety risks.

Description of the Proposed System

The proposed system begins by monitoring the operating conditions of the four-cell battery pack. Cell voltages are compared to determine whether a significant imbalance exists. Charging current, discharging current, and temperature are also monitored to assess the operating condition of the battery. The acquired parameters are passed to the preprocessing stage. The prepared input data are then supplied to the XGBoost model, which estimates SoH and RUL. SoH represents the battery's current condition relative to its expected condition when new, while RUL estimates the remaining useful service or cycling life according to the model's training and prediction definition.

4.1.2 Circuit Diagram of the DPP-Based Cell-Balancing System

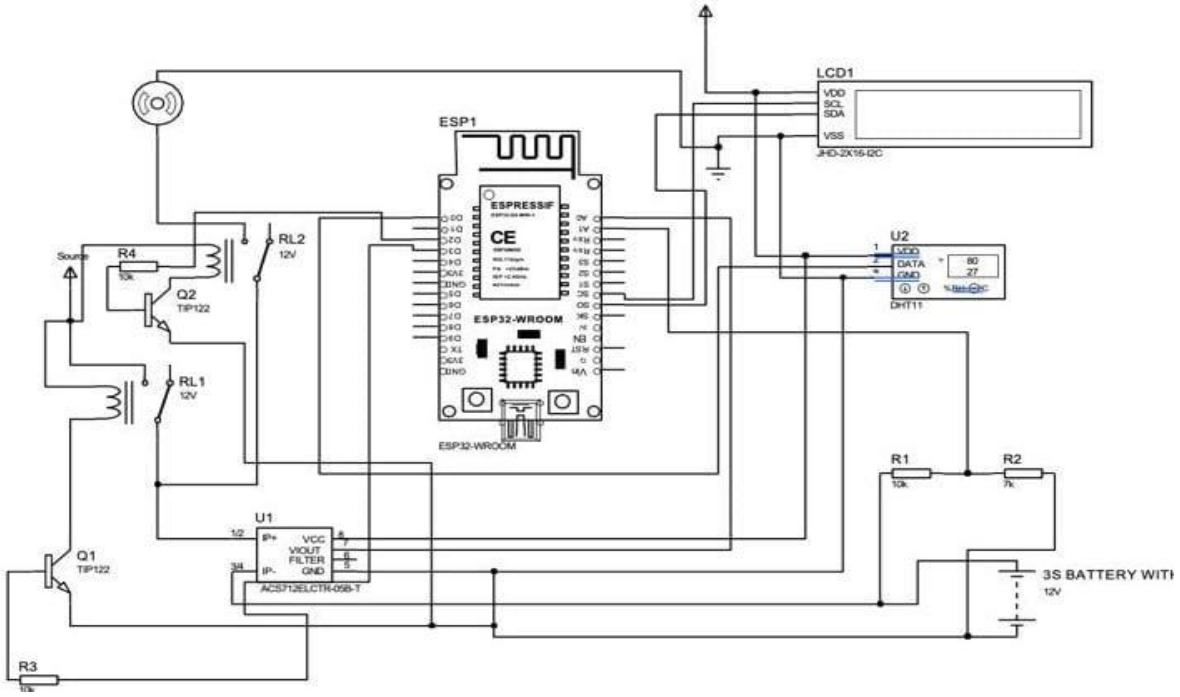


Figure 4.1 Circuit Diagram

The figure 4.1 circuit diagram should show the four battery cells, the three DPP converters, switching devices, inductors, relevant voltage and current measurement points, and the connections used for energy transfer. The exact component arrangement should match the circuit implemented in MATLAB/Simulink or the proposed hardware design.

Differential Power Processing is an active balancing approach in which the converter processes the difference in energy between cells rather than routing the entire battery-pack power through a balancing converter. In the proposed four-cell arrangement, the \(N-1\) topology uses three converters. This reduces the number of balancing converters compared with a separate-converter-per-cell arrangement, although the actual advantages depend on the circuit topology, component ratings, and operating conditions.

The converter inductors and switching devices determine the current-transfer behaviour. The switching frequency affects converter operation, while inductance influences the rate of current change and the amount of energy transferred during each switching interval. The proposed study considers a switching frequency of 50 kHz and inductance values of 100 μH, 10 μH, and 5.35 μH for examining balancing behaviour under the selected operating conditions.

4.1.3 Machine-Learning Algorithm for Battery Health Estimation

The machine-learning stage estimates battery health parameters from the selected operating data. XGBoost is used because it builds an ensemble of decision trees and combines their outputs to produce predictions. The method can represent nonlinear relationships between battery operating variables and battery health, provided that the training data adequately represent the conditions encountered during operation.

The input features considered in the proposed work are terminal voltage, charging current, discharging current, and temperature. The target outputs are SoH and RUL. The model requires appropriate labelled training data or reference values so that its predictions can be evaluated. Data preprocessing should ensure that the input features are represented consistently and that invalid or missing values are handled appropriately. The general algorithmic procedure consists of collecting battery data, preparing the input features, dividing the data into suitable training and evaluation sets, training the XGBoost model, and evaluating the predictions. Model parameters may be tuned using a validation procedure.

4.1.4 Data Acquisition and Preprocessing

Data acquisition is an important stage because the accuracy of the machine-learning model and the reliability of the charging controller depend on the quality of the input parameters. In the proposed system, the main parameters considered are battery terminal voltage, charging current, discharging current, and temperature. Individual cell voltage and SoC values are also monitored to evaluate the balancing operation. The intelligent charging controller combines battery health estimation, cell-balancing requirements, and safety checks to select the appropriate operating mode. The proposed controller considers four modes.

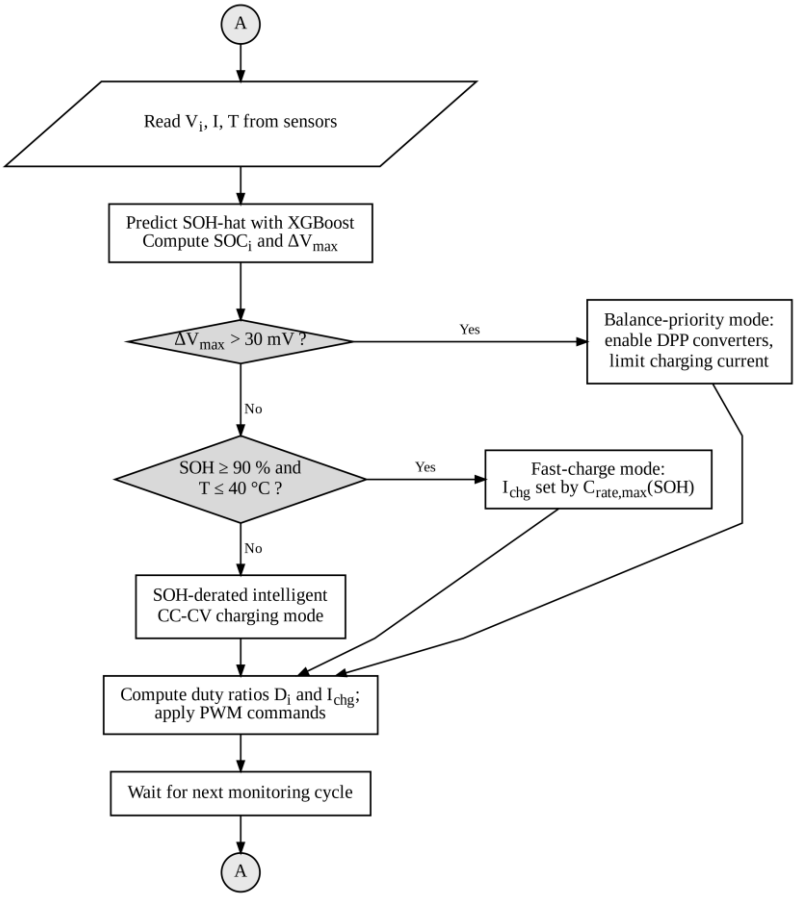


Figure 4.2 Flowchart of the ML-based intelligent and fast charging-mode decision executed each monitoring cycle

Intelligent Charging Control and Protection

Balance-priority mode: This mode is considered when the maximum difference between cell voltages exceeds 30 mV. The DPP converters are controlled to reduce the mismatch between cells. The controller continues monitoring the cell voltages to assess whether the imbalance is decreasing. SoH-derated CC-CV mode: This mode uses constant-current and constant-voltage charging principles while considering the estimated battery health. The charging current can be adjusted according to the defined SoH-based control strategy.

4.2 Methodology of the Proposed Work

The methodology integrates battery modelling, data acquisition, machine-learning-based health estimation, DPP cell balancing, charging-mode selection, and simulation-based evaluation. Each stage is developed and evaluated before the complete system is analysed. The following subsections describe the procedure used in the proposed work.

Battery Pack Configuration and Initial Conditions

The first stage is the development of a four-cell lithium-ion battery model in MATLAB/Simulink. The initial SoC values are set to 88%, 84%, 81%, and 78%. These values represent a battery pack with unequal charge levels and provide a basis for examining the cell-balancing response. The difference between the maximum and minimum initial SoC values is 10 percentage points. This initial spread is used as a reference for evaluating the effectiveness of the balancing process. The battery model provides the cell voltages and SoC values required by the control system.

These include terminal voltage, charging current, discharging current, and temperature. The selected parameter values and operating limits must be consistent with the battery model and the assumptions used in the study.

4.2.1 Data Processing and XGBoost Model Development

The second stage prepares the battery data for machine-learning-based health estimation. The input variables consist of terminal voltage, charging current, discharging current, and temperature. The output variables are SoH and RUL.

The dataset is checked for missing or invalid values, and the input features are arranged consistently. The data are then separated into training and evaluation sets according to the selected validation procedure. The XGBoost model is trained to learn the relationship between the input battery parameters and the target health values.

After training, the model generates SoH and RUL estimates for the evaluation data. The estimates are compared with reference values wherever they are available. Prediction error can be measured using metrics such as mean absolute error (MAE) and root mean squared error (RMSE). These metrics help determine how closely the model predictions agree with the reference values. Once the model has been evaluated, its outputs are connected to the charging controller. The estimated SoH supports the selection of charging conditions, while RUL provides an indication of the predicted remaining useful life. The model's suitability depends on the quality of the training data and its ability to represent the battery operating conditions being evaluated.

DPP Converter Design and Parameter Selection

The third stage develops the DPP-based cell-balancing subsystem. For four series-connected cells, the proposed \(N-1\) configuration uses three converters. These converters establish energy-transfer paths that allow the system to redistribute energy between cells.

The design considers a switching frequency of 50 kHz and inductance values of 100 μH, 10 μH, and 5.35 μH. A duty ratio of 0.30 is used in the specified example operating condition. These parameters are used to analyse the balancing current and the expected balancing behaviour.

The inductance value influences the rate of current change through the converter. A lower inductance can produce a higher current under otherwise comparable

operating conditions, but it also increases the need to check current ripple, component ratings, and switching losses. Therefore, the inductance should not be selected solely to maximise balancing current.

The converter model is integrated with the battery pack in MATLAB/Simulink. The switching and control signals are configured according to the selected converter topology. The resulting current and cell-voltage responses are recorded to evaluate whether the converter arrangement supports the required balancing behaviour.

Cell-Imbalance Detection and Balancing Operation

The fourth stage determines when balancing should be activated. The controller monitors the voltage of each cell and identifies the maximum and minimum cell voltages. The difference between these values is used as an indicator of cell imbalance.

When the voltage difference exceeds the selected threshold of 30 mV, balance-priority operation is considered. The DPP converters are then controlled to transfer energy through the available converter paths. This operation is intended to reduce differences between the cells and improve the usable capacity of the battery pack.

The balancing response is evaluated by recording individual cell voltages and SoC values over time. The reduction in the maximum-to-minimum SoC spread provides one measure of balancing performance. Balancing current and the time taken to reduce the mismatch are also relevant measures.

4.2.2 Integrated MATLAB/Simulink Simulation

After developing the battery model, machine-learning model, DPP converter subsystem, and charging controller, the individual blocks are integrated in MATLAB/Simulink. The purpose of this stage is to evaluate the combined operation of the proposed system under the selected simulation conditions.

The battery block provides cell voltage and SoC information. The data-processing and estimation blocks provide SoH and RUL estimates. The charging controller uses the health estimates and operating limits to select the charging mode, while the DPP control block regulates the energy-transfer process.

The integrated simulation is also used to check the coordination between the balancing controller and charging controller. For example, the balancing mode should respond to a voltage difference that exceeds the selected threshold, while protection conditions should override normal charging. The results are then compared with the expected behaviour of the model.

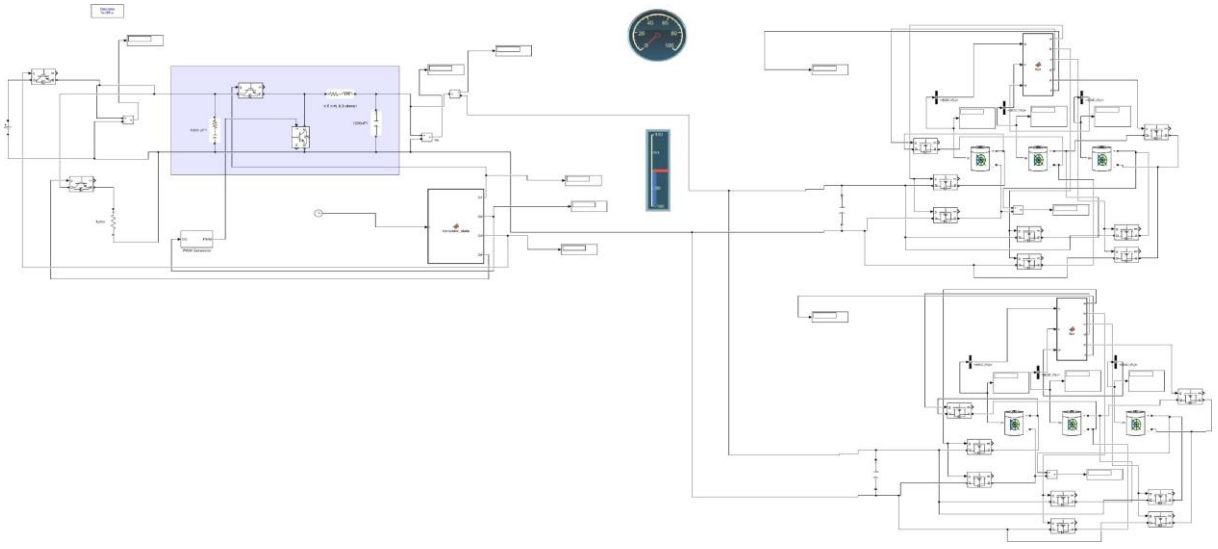


Figure 4.3 Simulation diagram

The simulation records the main output parameters, including individual cell SoC, cell-voltage difference, balancing current, charging current, temperature, and battery-health estimates. These outputs are analysed to determine whether the system responds to cell imbalance and changes in the battery operating condition as intended.

Simulation Findings and Performance Evaluation

The reported simulation results indicate a reduction in the difference between the cell SoC values during the balancing process. The initial SoC spread is 10

percentage points. It decreases to approximately 5 percentage points after 27 minutes, 2 percentage points after 110 minutes, and 1.03 percentage points after 240 minutes. The balancing-current calculation also illustrates the influence of converter inductance. Under the example conditions of a high-cell voltage of 3.70 V, a low-cell voltage of 3.60 V, switching frequency of 50 kHz, and duty ratio of 0.30, the reported balancing current is approximately 0.034 A for 100 μH and 0.64 A for 5.35 μH. These values illustrate the relationship between the selected inductance and the calculated current under the stated assumptions.

4.2.3 Limitations and Practical Considerations

The proposed system is evaluated primarily through analytical calculations and numerical simulation. The performance of a physical implementation may differ because of converter losses, parasitic resistance, switching delays, sensor errors, communication delays, and temperature variations. The charging limits used in the proposed controller must also be matched to the battery manufacturer's specifications. The 30 mV balancing threshold, 90% SoH fast-charging threshold, and 40°C temperature limit are project-specific control conditions, not universal limits applicable to every lithium-ion battery.

Future practical validation should include converter efficiency measurement, temperature monitoring during balancing, hardware protection testing, comparison of estimated and reference battery-health values, and evaluation under different initial cell imbalances. Such testing would establish whether the proposed design can achieve the intended performance in an actual battery system.

Summary

Presented the proposed machine-learning-based intelligent EV battery charging system using Differential Power Processing, battery health estimation, and thermal-aware charging control. The overall block diagram and proposed circuit configuration were described, along with the roles of the battery model, monitoring parameters, XGBoost algorithm, DPP converters, and charging

controller.

The methodology included battery-pack configuration, data preprocessing, SoH and RUL estimation, converter parameter selection, cell-imbalance detection, charging-mode selection, and integrated MATLAB/Simulink simulation. The reported simulation findings showed a reduction in the cell SoC spread from 10 percentage points initially to approximately 1.03 percentage points after 240 minutes. The results demonstrate the intended balancing behaviour of the model and the potential for health-aware charging decisions. Nevertheless, physical hardware validation and further testing are required to establish practical efficiency, safety, and long-term battery performance.

CHAPTER 5
RESULTS AND DISCUSSION

5.1 RESULTS

The horizontal axis represents the actual RUL, while the vertical axis represents the predicted RUL. The red diagonal line indicates the ideal condition in which predicted values are equal to actual values. The blue data points represent the model's predictions for different RUL values.

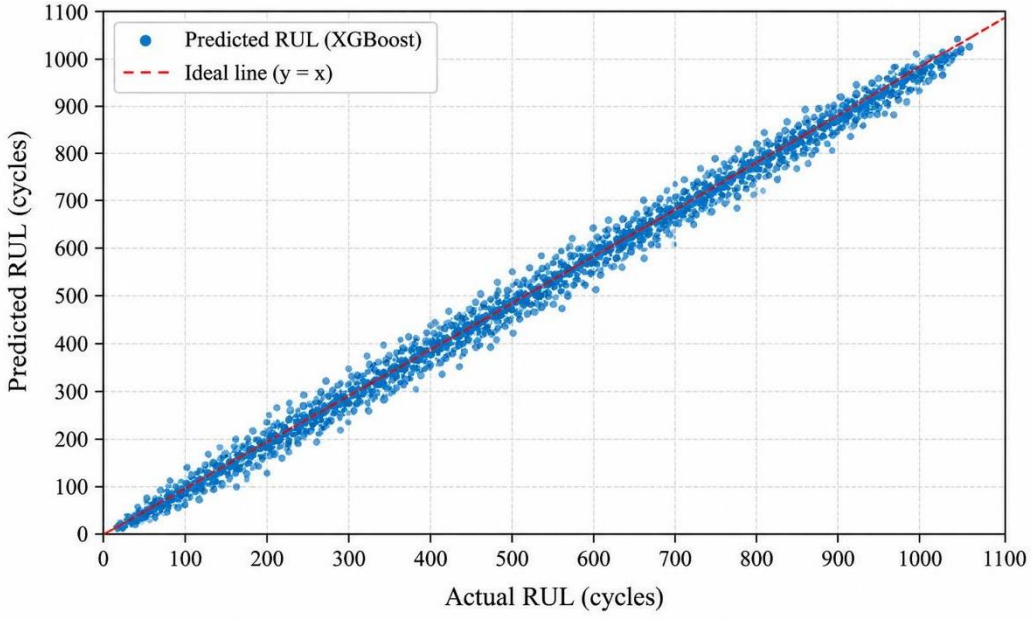


Figure 5.1 Actual vs. Predicted Remaining Useful Life (RUL) Using XGBoost

Figure 5.1 graph illustrates the performance of the Extreme Gradient Boosting (XGBoost) model in predicting the Remaining Useful Life (RUL) of a lithium-ion battery. Most data points are distributed closely around the diagonal line, indicating good agreement between the actual and predicted values. This demonstrates the model's ability to learn the relationship between battery operating parameters and remaining useful life. Accurate RUL prediction helps assess battery degradation and supports health-aware charging decisions. The estimated battery life can be used alongside voltage, current, and temperature. Voltage has the highest feature importance, approximately 0.58, followed by charging current at approximately 0.41. Temperature and discharging current show comparatively smaller contributions in the displayed model.

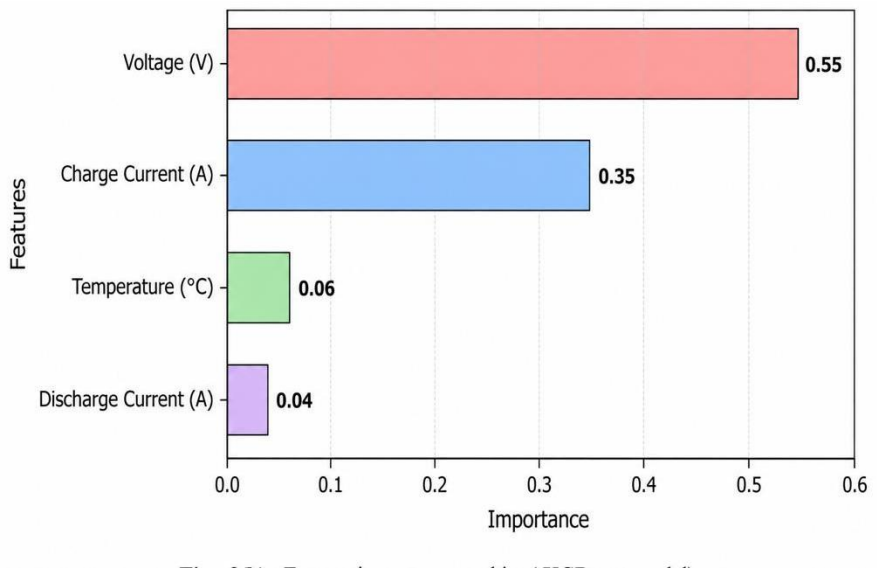


Figure 5.2 Feature Importance of XGBoost Model

Figure 5.2 graph illustrates the feature importance of the XGBoost model used for battery health estimation. It compares the contribution of four input parameters: voltage, charging current, temperature, and discharging current. These results indicate that voltage and charging current play the major roles in the model's predictions of battery health and Remaining Useful Life (RUL). Identifying the most influential parameters helps understand how the machine learning model uses battery operating data to generate predictions. The estimated battery health can support intelligent charging-mode selection and charging-current regulation. However, feature importance represents the relative contribution of input variables within the trained model and does not necessarily indicate a direct causal relationship. These findings can help prioritize battery parameter monitoring in the intelligent EV charging system.

The horizontal axis represents the duty ratio, ranging from 0 to 0.5, while the vertical axis represents the balancing current in amperes. The results indicate that the balancing current increases nonlinearly as the duty ratio increases. For a given duty ratio, a lower inductance produces a higher balancing current, whereas a higher inductance results in a lower balancing current. Among the four inductance values considered, 5.35 µH produces the highest current, followed by 10 µH, 22 µH, and 100 µH. At a duty ratio of approximately 0.5, the balancing current reaches about 1.75 A for 5.35 µH, while it remains around 0.1 A for 100 µH.

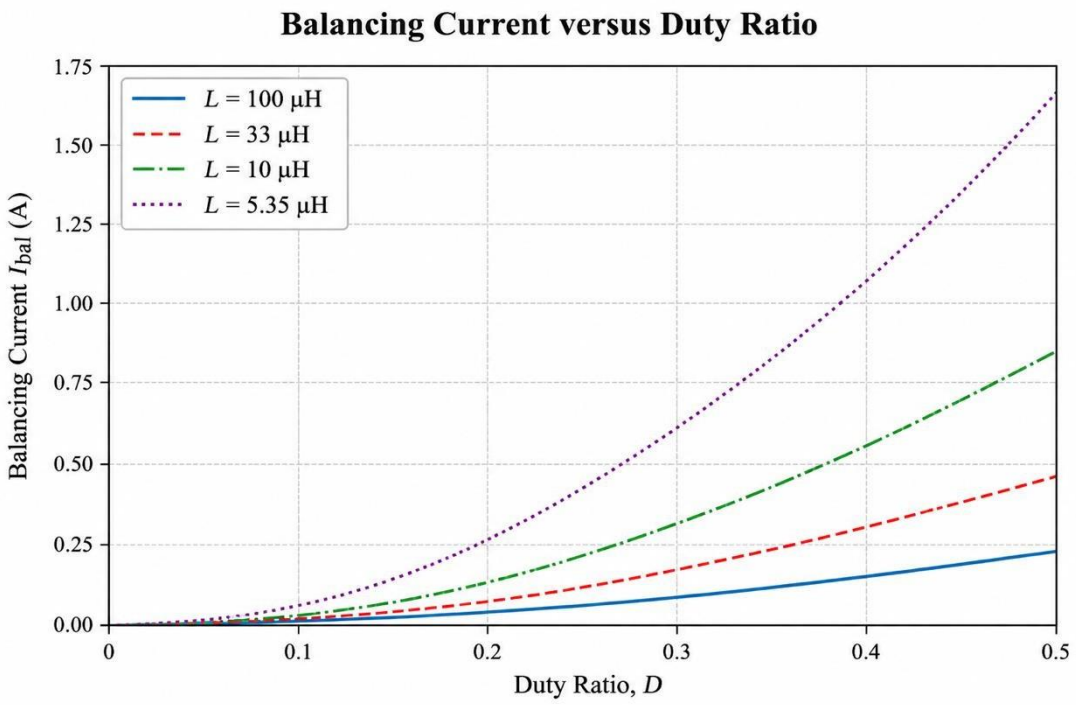


Figure 5.3. Effect of Duty Ratio on DPP Cell-Balancing Current

Figure 5.3 graph illustrates the relationship between the duty ratio and the balancing current for different inductance values in the Differential Power Processing (DPP) converter. The inductance value also significantly influences the balancing current. A higher balancing current can accelerate energy transfer between adjacent battery cells and reduce the time required to minimize State of Charge (SoC) differences. However, the inductance and duty ratio must be selected carefully to avoid excessive current and ensure reliable converter operation. The left plot shows the SoC variation of individual cells, initially set at different levels. As the balancing process progresses, energy is transferred between adjacent cells, causing their SoC values to gradually converge. Cell 1 starts with the highest SoC of approximately 88%, while Cell 4 begins at approximately 78%. The balancing process reduces these differences and brings the cell SoC values closer together. It decreases rapidly during the initial balancing period and gradually reduces as the cells approach equilibrium. The simulation results indicate that the SoC spread decreases to approximately 2% within 110 minutes and reaches nearly 1% after about 240 minutes.

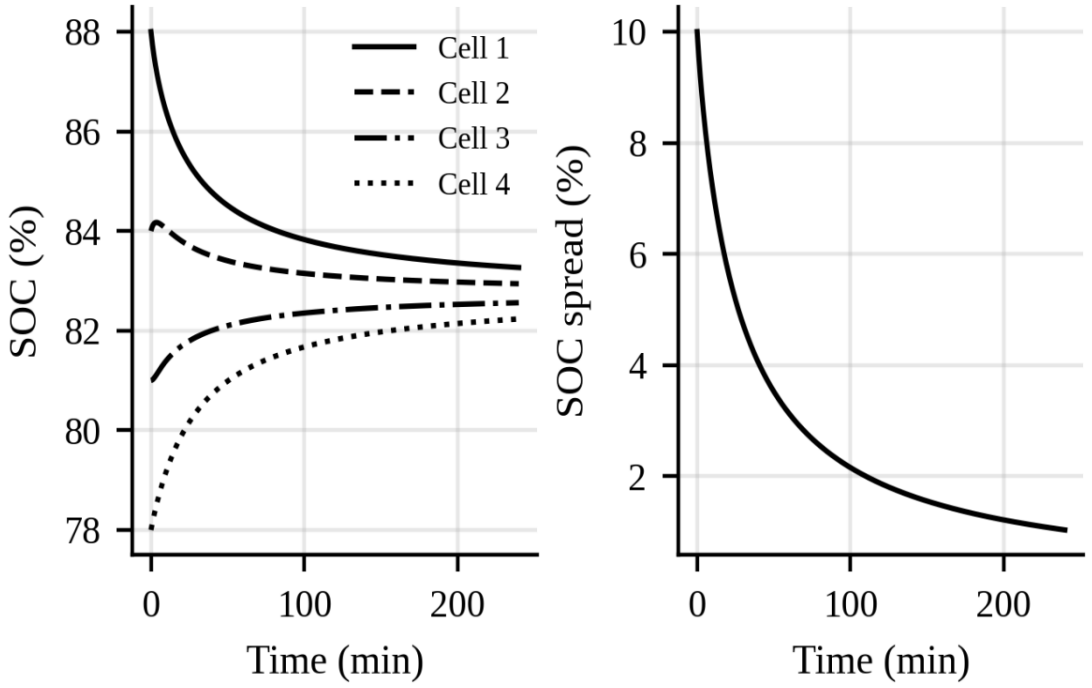


Figure 5.4 Variation of Cell SoC and SoC Spread During DPP Cell Balancing

Figure 5.4 graph illustrates the variation in the State of Charge (SoC) of four lithium-ion battery cells and the reduction in SoC spread over time during Differential Power Processing (DPP)-based cell balancing. The right plot shows the reduction in the overall SoC spread as balancing time increases. Initially, the SoC spread is approximately 10%. This behaviour demonstrates the effectiveness of DPP-based active cell balancing in series-connected battery cells. Improved cell balance can increase usable battery capacity and support more efficient. The left plot shows how the maximum charging C-rate varies with battery SoH for different exponent values ((n=1), (n=2), and (n=3)). As the SoH increases, the permitted charging C-rate increases, reaching approximately 2C at 100% SoH. When battery health deteriorates, the charging rate is reduced to limit stress on the battery. The graph indicates that the charging time for the fixed 2C strategy increases as SoH decreases, whereas the derated strategy maintains a comparatively consistent charging time by accounting for the reduced capacity of an aged battery.

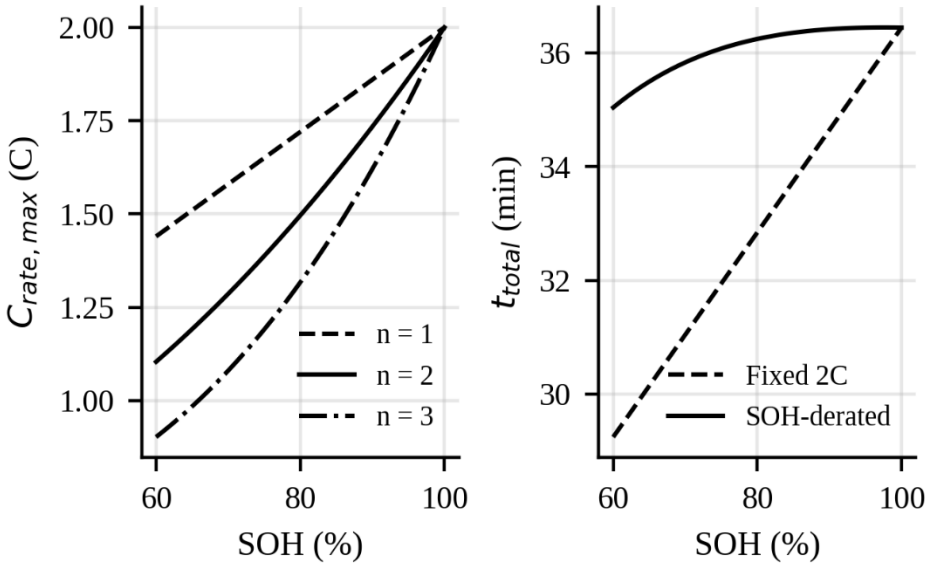


Figure 5.5 Effect of Battery SoH on Charging C-Rate and Total Charging Time

Figure 5.5 The graph illustrates the relationship between the State of Health (SoH) of a lithium-ion battery, the maximum permitted charging C-rate, and the total charging time. The rate of reduction depends on the selected exponent, which determines how strongly charging current responds to changes in SoH.

7.2 Significance, Strengths and Limitations of the Proposed Work

The proposed machine-learning-based intelligent EV battery charging system combines Differential Power Processing (DPP), battery health estimation, and intelligent charging control to improve battery management. PWM signals represent the converter switching commands. The control logic determines when balancing is required and provides the relevant command to the converter model. The direction of energy transfer must match the topology and the relative energy levels of the connected cells.

1. Significance and Strengths

The main significance of the proposed system is its ability to reduce cell imbalance while considering battery health and operating temperature. The DPP-based balancing method transfers energy between cells instead of dissipating excess energy entirely as heat. The XGBoost algorithm estimates State of Health (SoH) and Remaining Useful Life (RUL) using battery voltage, charging current,

discharging current, and temperature.

The simulation results show that the initial SoC difference of 10 percentage points decreases to approximately 1.03 percentage points after 240 minutes. This indicates improved cell balancing under the simulated conditions. The intelligent charging controller supports four operating modes: balance-priority, SoH-derated CC-CV, fast charging, and protection and monitoring. PWM signals represent the converter switching commands. The control logic determines when balancing is required and provides the relevant command to the converter model. The direction of energy transfer must match the topology and the relative energy levels of the connected cells.

These modes help adapt charging decisions to battery condition and safety requirements. MATLAB/Simulink also enables evaluation of the integrated system before practical implementation.

2. Limitations

The proposed system has certain limitations. First, the reported findings are based on analytical calculations and numerical simulation; physical hardware validation has not been established. PWM signals represent the converter switching commands. The control logic determines when balancing is required and provides the relevant command to the converter model. The direction of energy transfer must match the topology and the relative energy levels of the connected cells.

Second, the accuracy of SoH and RUL estimation depends on the quality and representativeness of the training data. PWM signals represent the converter switching commands. The control logic determines when balancing is required and provides the relevant command to the converter model. The direction of energy transfer must match the topology and the relative energy levels of the connected cells. Third, DPP converters introduce additional components, control complexity, and switching losses. The present study also considers a four-cell battery configuration, so scalability to larger EV battery packs requires further investigation.

5.2.1 COMPARISON OF EXISTING AND PROPOSED

Table 5.1 Comparison of Existing and Proposed System

S.NO	Parameter	Existing System	Proposed System
1	Charging Control Strategy	Conventional fixed CC–CV charging	Adaptive charging based on battery SOH
2	Cell Balancing Method	Passive balancing using resistors	Differential power processing (DPP) using bidirectional buck–boost converters
3	Battery Health Estimation	Limited or conventional estimation methods	XGBoost-based SOH and RUL prediction
4	Balancing Efficiency	Energy dissipated as heat	Mismatch energy transferred between cells
5	Charging Current Control	Fixed or predefined current limits	SOH- and temperature-dependent current control
6	Cell Voltage Balancing	Slower balancing with energy loss	Faster balancing through differential power transfer
7	Battery Life Management	Limited predictive capability	Health-aware charging and balancing
8	Fault Detection	Threshold-based monitoring	Sensor-based monitoring with predictive health information
9	System Monitoring	Basic voltage, current and temperature monitoring	Multi-parameter monitoring with an ESP32-based controller
10	Communication	Limited remote monitoring	Wi-Fi-based monitoring dashboard

5.3 COST BENEFIT ANALYSIS

The proposed intelligent EV battery charging system combines Differential Power Processing (DPP), machine-learning-based battery health estimation, and intelligent charging control to improve battery performance and energy utilisation. The cost of the system mainly depends on the battery pack, DPP converter components, sensors, ESP32 microcontroller, charging circuit, and monitoring devices. Additional costs are associated with software development, simulation, and system integration.

The ESP32-based monitoring arrangement provides a cost-effective option for processing battery measurements and supporting remote monitoring through Wi-Fi. MATLAB/Simulink enables system modelling and simulation before hardware implementation, reducing the need for repeated physical testing during the initial development stage.

Table 5.2: Cost Factors and Expected Benefits of the Proposed Intelligent EV Battery Charging System

Cost Factors	Expected Benefits
DPP converter components	Improved energy transfer between battery cells
Voltage, current, and temperature sensors	Continuous battery parameter monitoring
ESP32 microcontroller	Cost-effective control and data process
XGBoost algorithm	Battery SoH and RUL estimation
Wi-Fi monitoring dashboard	Remote monitoring of battery conditions
MATLAB/Simulink software	Simulation-based evaluation before hardware implementation

Table 5.3: Estimated Component-Wise Cost of the Proposed Intelligent EV Battery Charging System

S.No	Component	Quantity	Estimated Unit Cost (₹)	Total Cost (₹)
1	Lithium-ion battery cells	4	500	2,000
2	DPP converter components (MOSFETs, inductors, capacitors)	3 sets	300	900
3	Voltage sensors	4	75	300
4	Current sensor	1	200	200
5	Temperature sensors	4	50	200
6	ESP32 microcontroller	1	450	450
7	LCD display	1	180	180
8	Connecting wires and protection components	1 set	400	400
9	PCB and other accessories	1 set	300	300
	Total Estimated Cost			₹4,930

Summary

The proposed machine-learning-based intelligent EV battery charging system integrates Differential Power Processing (DPP), XGBoost-based State of Health (SoH) and Remaining Useful Life (RUL) estimation, and intelligent charging control. The simulation results demonstrate a reduction in the cell State of Charge (SoC) difference from an initial 10 percentage points to approximately

1.03 percentage points after 240 minutes. The system incorporates active cell balancing, health-aware charging modes, thermal monitoring, and protection mechanisms to improve battery management. Its major strengths include improved cell balancing and adaptive charging decisions based on battery operating conditions. However, the system has limitations related to training data accuracy, converter complexity, and the absence of practical hardware validation. Overall, the proposed approach shows potential for improving EV battery management, but further experimental testing is required to verify its efficiency, reliability, safety, and performance under real-world operating conditions. The cost-benefit analysis highlights the potential advantages of active balancing, ESP32-based control, and machine-learning-based health estimation. However, converter complexity, training-data accuracy, implementation cost, and the absence of hardware validation remain limitations. Overall, the simulation demonstrates the potential of the proposed approach, while experimental testing is required to verify practical efficiency, reliability, and safety.

CHAPTER 6

CONCLUSION AND FUTURE WORK

The proposed machine-learning-based intelligent EV battery charging system integrates Differential Power Processing (DPP), battery health estimation, and intelligent charging control to improve battery management. The system combines cell balancing with XGBoost-based State of Health (SoH) and Remaining Useful Life (RUL) estimation, while considering temperature and operating conditions for charging-mode selection. The simulation findings indicate a reduction in cell imbalance and demonstrate the potential of integrating machine learning with active battery balancing.

6.1 Conclusion

The proposed work focuses on developing a machine-learning-based intelligent EV battery charging system using Differential Power Processing for cell balancing and battery health management. A four-cell lithium-ion battery model with initial State of Charge (SoC) values of 88%, 84%, 81%, and 78% was considered. The system incorporates three DPP converters using an \(N-1\) configuration, with a switching frequency of 50 kHz and selected inductance values of 100 μH, 10 μH, and 5.35 μH.

The XGBoost algorithm is used to estimate SoH and RUL based on terminal voltage, charging current, discharging current, and temperature. The intelligent charging controller selects among balance-priority, SoH-derated CC-CV, fast-charging, and protection modes according to the defined operating conditions.

The simulation results indicate that the initial SoC spread of 10 percentage points decreases to approximately 5 percentage points after 27 minutes, 2 percentage points after 110 minutes, and 1.03 percentage points after 240 minutes. These findings demonstrate the ability of the simulated DPP system to reduce cell imbalance under the selected conditions. The proposed charging strategy also provides a framework for incorporating battery health and thermal constraints

into charging decisions.

Overall, the proposed approach demonstrates the potential of combining active cell balancing and machine-learning-based battery health estimation for EV battery management. However, the reported findings are based on analytical calculations and numerical simulations.

6.2 Suggestions for Future Work

The proposed system can be extended in the following ways to improve its practical performance and reliability.

1. **Hardware implementation**

   Develop a physical prototype using an ESP32 controller, voltage and current sensors, temperature sensors, and DPP converters to validate the simulation results.

2. **Advanced machine-learning models**

   Compare XGBoost with other machine-learning algorithms to improve SoH and RUL estimation accuracy using experimental battery datasets.

3. **Thermal management enhancement**

   Integrate a dedicated thermal management system to maintain battery temperature within manufacturer-specified limits during charging and balancing.

4. **Converter optimisation**

   Investigate converter efficiency, switching losses, inductance selection, and balancing speed to improve DPP performance.

5. **Larger battery configurations**

   Extend the four-cell model to a larger battery pack to evaluate scalability and the complexity of converter coordination.

6. **Real-time monitoring**

   Implement the proposed Wi-Fi dashboard to display cell voltage, SoC, SoH, temperature, charging current, and fault status.

7. **Experimental validation**

   Compare simulated and measured results under different initial SoC

conditions, temperatures, and charging rates to assess the robustness of the proposed system.

8. Economic and long-term evaluation

1.Measure practical energy losses, maintenance requirements, and battery degradation over extended operating cycles to determine the actual economic benefits.

2.Future development in these areas would help establish the practical feasibility of the proposed system and improve its suitability for real-world electric vehicle applications.

Summary

The proposed system integrates DPP-based cell balancing, XGBoost-based battery health estimation, and intelligent charging control for EV battery management. The simulation results demonstrate a reduction in cell SoC imbalance from an initial 10 percentage points to approximately 1.03 percentage points after 240 minutes. The study highlights the potential of combining active cell balancing with health-aware charging strategies. Future work includes hardware implementation, experimental validation, converter optimisation, and real-time monitoring to evaluate the system's efficiency, safety, reliability, and long-term performance under practical operating conditions.

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